Zhangir Nurmukhambetov

Engine and Tools Programmer
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Professional Experience

Tools Programmer April 2026 – Present
Flashgroup XR *Breda, Netherlands*

- Developed a custom Unity Collider Editor that lets artists and programmers create physics colliders directly in the Scene View by selecting mesh vertices, previewing collider shapes, and generating boxes, spheres, capsules, cylinders, convex hulls, or CoACD-based convex decompositions. Iterated on the tool through artist feedback, simplifying the UI and improving workflow speed, visual feedback, undo support, and generated asset persistence.

Tools Programmer November 2025 – April 2026
SportsImproVR *Breda, Netherlands*

- Delivered a full-body tracking prototype for Meta Quest by recreating VRChat's OSC-based tracking pipeline in Unity, enabling hip and leg movement to drive a VR tennis avatar. Researched existing tracking solutions, implemented an OSC receiver for Shiftall VR Manager data, and mapped tracker targets to a humanoid character using Unity IK.
- Built a Unity Editor one click video to FBX conversion pipeline by integrating MoveAI via a custom Python bridge, adding per clip progress and background execution, and enabling parallel conversions that cut a typical 3 video turnaround from about 15 minutes to about 5 minutes while removing manual upload, download, import steps.

Projects

Custom Voxel-Based Engine Puzzle Platformer | *C++, FMOD* May 2025 to Jun 2025

- Built a Unity style Particle System using EnTT ECS, cutting frame drops by 60%.
- Integrated 20+ FMOD sound events (e.g., footsteps, tongue, environment) for immersive feedback.
- Resolved 30+ gameplay bugs and led the Itch.io release, including trailer content and performance specs.

Custom Strategy Game Engine | *C++, OpenGL, GLSL, EnTT, ImGui, Cereal* Feb 2025 to Apr 2025

- Built a real time terrain editor with brush tools and prefab support, used across 100% of game maps.
- Implemented GPU based terrain editing and tessellation rendering, reducing latency by 90%.
- Created JSON based scene serialisation with EnTT Meta, halving iteration time on level testing.

IgKnighted | *Unreal Engine 5, C++, Blueprints, Gameplay Framework* May 2024 to Jun 2024

- Architected modular gameplay systems using Unreal's GameMode, GameState, PlayerState, and Actor Components to manage combat, player lifecycle, and match progression.
- Implemented local multiplayer with Enhanced Input, controller possession, respawning, and reusable combat abilities including shooting, dashing, and power-ups.
- Developed AI combat behaviors using Behavior Trees and Blackboards, alongside data-driven enemy encounters and dynamic difficulty scaling.

Technical Skills

Languages: C++, C#, GLSL, HLSL, Python
Frameworks & Libraries: OpenGL, EnTT (ECS), ImGui, Cereal, GLM, tinygltf, ImGuizmo
Game Engines: Unreal Engine 5, Unity (C#), Godot
Developer Tools: Visual Studio, Git, Perforce (P4V), GitHub Actions
Languages: English (Fluent), Russian (Fluent), Arabic (Beginner)
Interests: Creative writing, Digital art and design

Education

Breda University of Applied Sciences September 2023 to Expected Jun 2026
Bachelor of Science in Creative Media and Game Technologies *Breda, Netherlands*